

Training for the Future: Preparing Humans for Effective Human-AI Programming

Qianou Ma

Human-Computer-Interaction Institute
Carnegie Mellon University
Pittsburgh, Pennsylvania, USA
qianoum@cs.cmu.edu

Abstract

Large language models are reshaping what it means to program, yet people benefit from them unevenly and often struggle to collaborate with AI effectively. My dissertation investigates how to prepare humans — especially non-experts — for productive human-AI programming by identifying key human skills, designing systems that teach those skills, and developing methods to evaluate learning and collaboration outcomes. Drawing on HCI and the Learning Sciences literature, I study human behavior in AI-assisted workflows, design LLM-based learning tools that scaffold communication and evaluation, and propose frameworks for assessing human-AI interaction beyond model accuracy. Through classroom studies, controlled experiments, and system deployments, my work aims to ensure that increasingly capable AI systems strengthen, rather than erode, human reasoning and agency. This dissertation contributes empirical insights, pedagogical designs, and evaluation approaches to support effective and equitable human-AI collaboration, particularly within the programming domain.

CCS Concepts

- **Human-centered computing** → **Interactive systems and tools**; • **Applied computing** → **Computer-assisted instruction**;
- **Computing methodologies** → Natural language generation.

Keywords

Human-AI Interaction, Programming, Learning Sciences, Computing Education

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1 Dissertation Scope and Planned Contributions

Large language models (LLMs) are redefining what it means to program and broadening who can develop software apps or analyze complicated data — from CS experts to non-programmers, without writing much code. Yet, both my work [2] and others' [7] reveal



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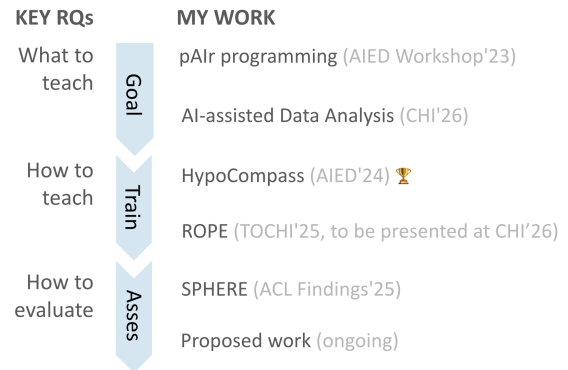


Figure 1: My research examines key AI literacy goals, effective tutoring, and how to evaluate learning and human-AI systems in AI-assisted programming workflows.

a gap between this vision and reality: people benefit non-equally from LLMs, and AI's impact depends not only on algorithmic power but on how humans collaborate with them. My research, therefore, asks: **How can we enable everyone, even the non-experts, to program productively with AI and solve complex tasks?** How should we best prepare humans for the AI era? To this end, my dissertation consists of three interrelated aims or core research questions (Fig. 1):

- (1) **What should humans still learn** to collaborate effectively with AI programming assistants?
- (2) **How can we best teach humans** to develop skills for effective human-AI programming?
- (3) **How do we evaluate human-AI** system effectiveness and human learning?

1.1 Skills to teach: communication and evaluation as enduring human skills.

I identify necessary human skills for future “programming” via both empirical study and literature review. On one hand, I conducted a semester-long classroom study to understand challenges in AI-infused workflows [2]. I analyzed over 7,000 logs from 36 students performing AI-assisted data analysis in computational notebooks, and using statistical modeling and other mixed-methods analysis, I distilled expert vs. novice behavior differences and homework grades progressions. On the other hand, I reflected on whether “AI pair programmers” actually mirror human partners, drawing on

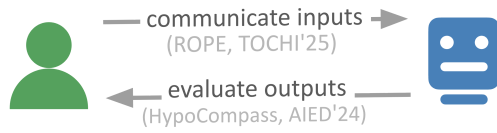


Figure 2: Key AI literacy skills of evaluation and communication for humans to learn in the AI era.

human–human pair-programming literature to surface skills that transfer to human–AI work [5].

Both studies highlight an important reorientation: humans need to **prioritize the higher-order thinking skills of communication and evaluation** to co-work with AI, which are **enduring human skills regardless of model capacity**. As models can only partially infer users’ intent from underspecified prompts [6], humans must express needs clearly, especially for complex and customized tasks like analyzing unique datasets. And because humans remain accountable for AI outputs, they must evaluate systematically and overcome challenges like expertise mismatch and cognitive overload, as revealed in my studies. However, my classroom study exposed a critical gap in *AI literacy instructions*: demonstrations and lectures fail to teach how to *prompt* and *evaluate* model behavior effectively. For example, students may fail to accurately describe customized visual elements they want, resulting in a wrong chart; or students may not check for all criteria for good visualization, submitting suboptimal plot [2]. These works grounded my design thread of work: if we can characterize these key skills, we can *intentionally design systems to scaffold them*.

1.2 Ways to teach: appropriate task-delegation for deliberate practice.

Building on the above findings, I design and build systems that explicitly train on two higher-order skills central to human–AI programming: evaluation and communication (Fig. 2).

To help end-users **better express their needs in prompts**, I developed a suite of prompt programming tasks (e.g., replicate customized LLM apps), assessments, and a LLM-based tutoring system, ROPE [3]. Building on a well-established field of requirement engineering in CS, ROPE (Fig. 3) proposes an emerging task delegation strategy between humans (who clearly specify requirements) and AIs (that compile optimized prompts). Controlled experiments revealed that end-users trained with ROPE doubled their prompting performance and produced significantly more reliable model outputs, compared to conventional prompt engineering training.

To prepare students to **better think about failure cases in AI outputs**, I built HypoCompass, an LLM-based teachable agent where novice CS students act as tutors to diagnose and explain the AI’s mistakes to AI “learners.” [4]. This teachable agent setup fosters appropriate reliance and presents AI bugs to human novices in an error-tolerant environment. Pre-post test showed that HypoCompass significantly improved debugging performance by 12% and efficiency by 14%, while generating practice materials four times faster than instructors. HypoCompass won the **Best Paper** at AIED 2024 (top-1).

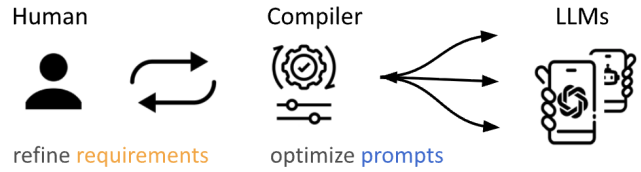


Figure 3: Envisioned ROPE paradigm [3]: requirement-oriented prompt engineering.

A key factor in both systems’ pedagogical effectiveness lies in their controlled AI behaviors. These systems present an **idealized task delegation** between human and AIs, envisioning future no/low-code or natural-language programming workflows. I design the learning interactions such that learners only need to practice elaborating requirements or hypotheses tests, while code generation and bug fixing are offloaded to highly scaffolded AIs, thus removing distractions from real-world AI limitations that often frustrate or confuse novices. In parallel, my other work exposes the general public to current AI flaws to help them reflect on model behaviors like bias [1]. Together, my learning systems illustrate how imperfect AIs can be designed to **strengthen** human reasoning rather than replace or damage it.

1.3 Needs to evaluate: systematic and automated assessment.

Across my work, I observe a recurring methodological challenge in **evaluating both model utility and human learning outcomes**. In my studies, I consistently draw from NLP and HCI literature to include both intrinsic measures on the model performance and extrinsic evaluation tasks and rubrics on human–AI collaboration in specific contexts. I not only integrate common metrics from the pair programming literature, such as time on task and test coverage [5], but I also develop assessments for human learning, such as requirement quality in prompts [3], based on the requirement engineering literature.

In a prior work, I co-led the development of SPHERE [8], a framework that unifies quantitative evaluation from NLP evaluation with human-centered depth from HCI. We surveyed the literature and applied SPHERE to 39 published systems in both NLP and HCI, which surfaced current evaluation practices and gaps in validation and grounding. By formalizing the shared evaluation language, SPHERE helps researchers and human–AI system developers design and document their evaluation. In my ongoing work, I aim to more systematically and automatically assess humans’ workflow with current AI coding assistants, integrating the lessons learned from designing evaluations across my work.

2 Proposal and Future Work

Together, my dissertation establishes an integrated agenda for training humans to program with AI productively — from (1) diagnosing skill gaps in human–AI collaboration to (2) building systems that teach those skills, and finally, to (3) developing frameworks that measure their impact. I plan to expand this vision along two directions. First, I am building scalable assessment pipelines that

automatically evaluate human-AI programming workflows. Furthermore, I plan to generalize my lessons learned from training key technical literacy skills across different disciplines, like data science and quantitative reasoning, to encourage transfer. Ultimately, I am striving to ensure that as AI becomes more capable, humans are well-equipped to maximize its utilities and minimize the risks with the necessary skills.

2.1 Scalable assessment of human-agent programming workflows.

Following up on my (3) evaluation thread of work, I plan to develop scalable automatic assessment pipelines for evaluating human-agent programming workflows beyond final code correctness. Specifically, we deploy instrumented development environments that log fine-grained interaction traces between users and AI coding agents, including prompt histories, intermediate code states, and revision trajectories. Using these data, I will operationalize rubric-based and semi-automated metrics that assess higher-order human skills reflected in the development process, such as requirement specification, error recovery, and iterative testing behavior. Planned studies include longitudinal classroom deployments with open-ended course projects, which should reflect realistic, open-ended development workflows. We aim to examine whether our metrics can reliably differentiate levels of human-AI collaboration skill, remain meaningful as AI models change, and predict diverse downstream learning or performance outcomes. Through this project, I hope to establish assessment methods that scale to authentic educational and professional contexts without collapsing complex collaboration into binary notions of success or failure.

2.2 Generalizable AI literacy across domains.

Building on my (2) prior tutoring systems that scaffold communication and evaluation, I plan to extend this assessment-and-training framework beyond CS1 or application development to investigate how key technical literacy skills transfer across technical domains such as quantitative reasoning and modeling. I am developing an AI-based teachable agent that shifts instruction away from procedural computation and toward higher-order reasoning skills, including problem formulation, assumption checking, and solution evaluation. In this setting, students learn by authoring real-world modeling problems for an AI agent and critiquing its solutions using explicit rubrics, mirroring the communication and evaluation demands observed in AI-assisted programming. Planned classroom or crowd-sourcing studies will examine whether training these skills supports transfer to unfamiliar AI-assisted modeling tasks, or is better than training on procedural computation. By comparing learning trajectories across different computational domains, this work aims to identify which aspects of human-AI task delegation literacy generalize across technical disciplines and which require domain-specific scaffolding, contributing to a unified account of how humans can reason effectively with AI.

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